

by Taryn Plumb

Cloudflare wants to provide the operating system for the AI-first enterprise

News

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Cloudflare OS brings AI workspaces, zero-trust access, governed connectors, model routing, and enterprise data together in the browser, while new tools help IT teams track AI identities, token spending, and insider risk.



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Traditional operating systems (OS) were built to manage hardware, files, apps, and users on a device, but Cloudflare says the agentic AI era requires a whole new format.

The company this week announced **Cloudflare OS** [<https://www.cloudflare.com/press/press-releases/2026/cloudflare-os-is-the-first-ai-workspace-built-around-how-companies-actually-work/>], which connects AI agents, enterprise data and context, internal systems, and workflows together in one secure workspace. It is open source and browser-based, sparing companies the need to build all-new infrastructure.

The OS is launching alongside several other new security, identity, spending, and user insight tools that Cloudflare has built for the **AI-based workplace** [<https://www.infoworld.com/article/2255318/what-is-cloud-native-the-modern-way-to-develop-software.html>].

“Cloudflare OS isn’t a traditional desktop OS,” said **Rita Kozlov** [<https://www.linkedin.com/in/ritakozlov/>], VP of product at Cloudflare. “It reimagines the workplace computing environment for AI.”

Open source OS runs in a browser

Cloudflare OS serves as a secure, AI-equipped workspace that is plugged into internal company systems. Available now through Cloudflare’s open source repository, it is accessible directly in a browser, and runs inside an enterprise’s Cloudflare account.

“It is a browser-based workspace that begins with a conversation,” Kozlov explained. Users can ask an agent to research, create slides, spreadsheets, and documents, build full-stack apps, or automate workflows without the need for a terminal. Those outputs are then shareable, but kept in isolated databases with access controls.

Enterprises will soon be able to access the OS directly through Cloudflare or via a “select group” of partners that will build tailored offerings on Cloudflare’s architecture, the company says. Because it is open source, organizational processes, internal system connections, and context aren’t locked into a vendor product or AI model provider. Customers can use whatever models they choose.

Cloudflare OS is built on Cloudflare Workers, **Dynamic Workers** [<https://www.infoworld.com/article/4149869/cloudflare-launches-dynamic-workers-for-ai-agent-execution.html>], Durable Objects, and Access, the company’s zero trust network access (ZTNA) tool that verifies every user and request. Agents start with zero permissions by default and are only granted access to tools required for a specific task. Organizations configure their own Access policies, models, branding, skills, and integrations, Kozlov explained.

Governed connectors known as gatekeepers give admins control over what AI can see, what it can change, and when the system needs human sign-off. They can also control budgets, set rate limits, and delegate tasks to different models.

“Because agents act on people’s behalf [<https://www.infoworld.com/article/4165857/are-we-ready-to-give-ai-agents-the-keys-to-the-cloud-cloudflare-thinks-so.html>] and produce work others can access and modify, they require a new security model,” Kozlov said. Thus, Cloudflare OS tracks the resources an agent requires so the right access controls follow its work when it is shared.

Cloudflare initially built the OS for internal use, and employees “across every team” use it daily. Kozlov estimated that, over the last 30 days, internal users have used it to create more than 4,000 apps, automations, and tools. Over that same period, she claimed, the company’s sales team saved an estimated 10,000 hours by automating previously manual tasks like territory planning and proposal creation.

“We open sourced Cloudflare OS so any organization can build ‘Your Company OS,’” Kozlov said. Open source is critical because “you cannot put your company into software you do not own. Organizations need to be able to inspect the platform, customize it, connect their own systems, and make it their own,” she explained.

A more cohesive bundle

Cloudflare deserves credit for packaging Cloudflare OS as an operating system, noted tech analyst Carmi Levy [<https://ca.linkedin.com/in/carmi>].

“This very much is not Windows, macOS, or Linux, and it isn’t an operating system by its common definition,” he said. “But Cloudflare’s use of this terminology implies familiarity to enterprise IT buyers.”

This makes for an easier discussion as enterprises struggle to understand how to best incorporate AI-related platforms and workflows into infrastructure that wasn’t initially designed for it.

Microsoft has marketed the combination of its Azure, Entra, Fabric, Windows, and Microsoft 365 offerings as an operating system of sorts, but hasn’t pulled all the pieces into a common brand, Levy said. And Google’s Gemini, Workspace, Vertex AI, and Cloud Run are “circling similar territory.”

But, he noted, Cloudflare OS is “more cohesively bundled” and infrastructure-focused, offering a single pane of glass platform for buyers worried about stitching together otherwise disparate AI-aware networking pieces. The company recognizes that AI introduces new architectural realities such as inference and model routing “over and above” traditional OS core competencies.

“While competing offerings generally leave the infrastructure heavy lifting to enterprise decision-makers, Cloudflare is marketing itself as a single-source vendor, which

potentially frees IT planners from having to integrate all the AI pieces on their own,” Levy said.

An infrastructure-first, application-agnostic approach means Cloudflare OS can coexist with whatever AI applications already exist in an enterprise, he said. It will “play nice” with OpenAI, Anthropic, Google, Microsoft, Meta, or open source layers, allowing employees to begin working in familiar workflows after sign-in.

“Its open-source architecture also minimizes the potential for vendor lock-in as enterprises gradually figure out how to evolve their stacks to align with new AI-era realities,” Levy said.

Managing identities and budgets for both humans and AI

As AI agents emerge across the enterprise, tracking their use can be challenging, causing problems from both a security and a spend standpoint. Along with Cloudflare OS, the company has launched a way to address this issue with its new **Identity-Aware AI Gateway** [<https://www.cloudflare.com/press/press-releases/2026/cloudflare-gives-companies-full-visibility-to-audit-and-analyze-ai-use/>], now in beta.

Also integrated with Access, the offering gives admins visibility into what users (both human and AI) are requesting from AI models. It allows security teams to set up custom domains in front of their gateways and replace shared API keys by integrating with their identity provider, like Okta or Entra, and ZTNA infrastructure, Cloudflare explained.

Every request is tied to Access-verified identities, and enterprises can filter each user’s logs, analytics, and spend. IT teams can track redundancies, limit usage rates, and apply filters that strip out employee names, passwords, and other sensitive data before requests go to outside model providers.

A companion feature, AI Spend, tracks every user’s behavior over time to create a baseline of normal AI usage. When spending deviates from that pattern, the system alerts the IT team.

A new tab, User Insights, tracks cost and identifies over-spend caused by activities such as low cache-hit rates or oversized context windows. The capability scores sessions and compares them against account history using a 95th percentile session cost over the previous 30 days, Cloudflare product managers **Ming Lu** [<https://blog.cloudflare.com/author/ming-lu/>], **Kenny Johnson** [<https://blog.cloudflare.com/author/kenny/>], and **Ayush Kumar** [<https://blog.cloudflare.com/author/ayush/>] explain in a **blog post** [<https://blog.cloudflare.com/identity-aware-ai-gateway/>]. Anything above 2x an account’s 95th percentile is a “strong candidate for anomalous behavior.”

For instance, one Cloudflare customer had an employee who left a rogue AI session running, generating a \$30K bill. “User Insights helped them identify the problem and shut off access before the problem was further exacerbated,” Kozlov said.

Cloudflare is also building prompt classification functionality that sorts requests into categories such as coding or writing. This can help enterprises understand what AI is being used for.

“Once business traffic is separated from everything else, personal use becomes visible,” the project managers explained. “From the outside, someone running a side hustle on company time and someone quietly moving data out through a model look the same. Telling them apart is central to catching insider risk.”

Looking at the bigger picture

Identity-Aware AI Gateway and AI Spend address the visibility problem that has dogged so many recent AI deployments where enterprises failed to monitor usage, Levy noted. Projects “crashed and burned” as users unwittingly blew through token allocations.

These platforms provide single-point visibility into what is being used, how it’s being used, and where the potential lies for raising the productivity bar, he said. They overlay with existing models; in doing so, they enhance security with more precise control over resource allocations, and via automated anonymization protocols that prevent inadvertent sharing of sensitive data.

Ultimately, he said, vendors who free IT from having to independently assemble the pieces of their own AI implementations, and who assist them with answers to AI-specific questions, “will gain advantage over vendors that aren’t looking at the bigger picture.

*This article originally appeared on **CIO.com** [<https://www.cio.com/article/4206332/cloudflare-wants-to-provide-the-operating-system-for-the-ai-first-enterprise.html>].*

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